Equation sheet: governing relations of the $CaCO_3$-barrier screen

P0-W3.2 (cp2 fix-up), contract v3. Symbolic only; every symbol is in the table overleaf and in `theory/symbols.yaml`; units in brackets.

	R-slurry: aqueous, alkaline, room temperature; casting and drying on native Al_2O_3/AlOOH	R-cell: non-aqueous, fluorine-free electrolyte, collector at anodic potential vs Li^+/Li
Links that apply	C1 adhesion (1), C2 sequestration (2), supersaturation (3), C3 nucleation and Young (4, 5), template action (6), C4 morphology (7), C5 aqueous Pourbaix (8a)	C5 Li-open stability window (8b), C4 insulating-film bound (9). Not defined here: (1)–(7), since there is no hydrated Ca^{2+} reference and no aqueous Ω

1 Adhesion in water (C1, R-slurry)

The polymer never meets a dry oxide: every contact first displaces interfacial water, so the decisive work separates the contact into two water-wetted *surfaces.*

$$W_{\mathrm{adh,wet}} = \gamma_{\mathrm{ow}} + \gamma_{\mathrm{pw}} - \gamma_{\mathrm{op}} \quad [\mathrm{J\,m^{-2}}] \tag{1}$$
$$= W_{\mathrm{adh,dry}} + 2\gamma_{\mathrm{w}} - W_{\mathrm{ow}} - W_{\mathrm{pw}}$$
$$W_{\mathrm{adh,dry}} = \gamma_{\mathrm{o}} + \gamma_{\mathrm{p}} - \gamma_{\mathrm{op}} \quad [\mathrm{J\,m^{-2}}]$$
$$W_{\mathrm{adh,wet}} = \Delta F_{\mathrm{PMF}}/A_{\mathrm{c}} \quad [\mathrm{J\,m^{-2}}] \tag{2}$$

Anchored against water iff $W_{\mathrm{adh,wet}} > 0$; $W_{\mathrm{ow}}, W_{\mathrm{pw}}$ are the dry works of the oxide and polymer with water.

2 Sequestration (C2, R-slurry)

The template raises local Ca^{2+} only if it holds Ca^{2+} more strongly than water does; the cycle starts from hydrated Ca^{2+} at the regime activity.

Cycle: Ca^{2+}(aq, bulk, $a_{Ca^{2+}}$) $\rightarrow$ Ca^{2+}(binding layer). Solution side: molal standard state at the regime ionic strength; layer side: site-volume standard state, $\Delta G_{\mathrm{site}} = -RT\ln\big[V_{\mathrm{site}}^{-1}\int_{\mathrm{site}} \mathrm{e}^{-w(\mathbf{r})/k_{\mathrm{B}}T}\mathrm{d}^3r\big]$, with w the per-ion PMF ($w \rightarrow 0$ in bulk). With one coordinating group that must be deprotonated:

$$\Delta G_{\mathrm{seq}} = \Delta G_{\mathrm{site}} + RT\ln\big(1 + 10^{\,\mathrm{p}K_{\mathrm{a}} - \mathrm{pH}}\big) \quad [\mathrm{J\,mol^{-1}}] \tag{3}$$

$$\mathcal{E} \equiv \frac{c_{\mathrm{local}}}{c_{\mathrm{bulk}}} = \exp\Big(-\frac{\Delta G_{\mathrm{seq}}}{RT}\Big) \quad [1] \tag{4}$$

Valid for dilute Ca^{2+} with unsaturated sites ($\mathcal{E}c_{\mathrm{bulk}} \ll c_{\mathrm{max}}$). $\mathcal{E}$ is a *concentration* ratio: at equilibrium the Ca^{2+} activity in the layer equals the bulk one, and the enrichment sits in a local activity coefficient. Langmuir closure, with x the occupancy odds:

$$c_{\mathrm{local}} = \frac{\mathcal{E}c_{\mathrm{bulk}}}{1+x} = c_{\mathrm{max}}\Theta, \quad x = \frac{\mathcal{E}c_{\mathrm{bulk}}}{c_{\mathrm{max}}}, \quad \Theta = \frac{x}{1+x} \tag{5}$$
$$c_{\mathrm{max}} = \Gamma_{\mathrm{s}}/\delta_{\mathrm{s}} \ [\mathrm{mol\,m^{-3}}], \qquad \Gamma_{\mathrm{Ca}} = \Gamma_{\mathrm{s}}\Theta - \delta_{\mathrm{s}}c_{\mathrm{bulk}} \ [\mathrm{mol\,m^{-2}}]$$

Gibbs adsorption: binding lowers the template–liquid energy, $\mathrm{d}\gamma_{\mathrm{sl}} = -\Gamma_{\mathrm{Ca}}\,\mathrm{d}\mu_{\mathrm{Ca}}$*, integrated from zero to the regime activity:*

$$\gamma_{\mathrm{sl}} = \gamma_{\mathrm{sl}}^{(0)} + \Delta\gamma_{\mathrm{ads}} \quad [\mathrm{J\,m^{-2}}] \tag{6}$$
$$\Delta\gamma_{\mathrm{ads}} = -RT\big(\Gamma_{\mathrm{s}}\ln(1+x) - \delta_{\mathrm{s}}c_{\mathrm{bulk}}\big)$$
$$\partial\Delta\gamma_{\mathrm{ads}}/\partial\Delta G_{\mathrm{seq}} = \Gamma_{\mathrm{s}}\Theta > 0$$

$\Delta\gamma_{\mathrm{ads}} \equiv 0$ if γ_{sl} is computed at the regime activities. γ_{sn} is likewise grand-canonical: a Ca^{2+} excess at the contact (ion-bridged carboxylate–Ca–calcite) is inside it, so

$$\partial\gamma_{\mathrm{sn}}/\partial\mu_{\mathrm{Ca}} = -\Gamma_{\mathrm{Ca,sn}}, \qquad \partial(\gamma_{\mathrm{sl}}-\gamma_{\mathrm{sn}})/\partial\mu_{\mathrm{Ca}} = -(\Gamma_{\mathrm{Ca,sl}}-\Gamma_{\mathrm{Ca,sn}}) \tag{7}$$

with $\Gamma_{\mathrm{Ca,sl}} = \Gamma_{\mathrm{Ca}}$. Rule for supplied inputs: γ_{sn} comes from contact data at the regime state; ΔG_{seq} is never added to it (no claim that $\Gamma_{\mathrm{Ca,sn}} = 0$).

3 Supersaturation (one definition)

Crystallisation is driven by the excess chemical potential of the dissolved ion pair over the solid of the registered framing (ACC or calcite).

$$\Omega = \frac{\mathrm{IAP}}{K_{\mathrm{sp}}} = \frac{a_{Ca^{2+}}\,a_{CO_3^{2-}}}{K_{\mathrm{sp}}}, \qquad \Delta\mu = k_{\mathrm{B}}T\ln\Omega \quad [\mathrm{J}] \tag{8}$$

per $CaCO_3$ formula unit (V_{m} also per formula unit); nucleation requires $\Omega > 1$. **At the template $\Omega = \Omega_{\mathrm{bulk}}$.** At equilibrium $\tilde{\mu}_{\mathrm{Ca}}$ and $\tilde{\mu}_{CO_3}$ are uniform; a mean potential ψ gives factors $\mathrm{e}^{\mp 2e\psi/k_{\mathrm{B}}T}$ that cancel in the neutral solid, and specific binding lowers the local standard potential by exactly what it raises the concentration. Neither changes $\Delta\mu$:

$$\Delta\mu = \tilde{\mu}_{\mathrm{Ca}} + \tilde{\mu}_{CO_3} - \mu_{CaCO_3} = k_{\mathrm{B}}T\ln\Omega_{\mathrm{bulk}} \quad [\mathrm{J}] \tag{9}$$

Activities from speciation (pH, C_{T}, $\mathrm{Ca_T}$, I; pairs $CaCO_3^0$, $CaHCO_3^+$, $CaOH^+$; Davies, $I \le I_{\mathrm{val}}$):

$$a_{CO_3^{2-}} = \frac{2m_{C_{\mathrm{T}}}D_{\mathrm{c}}}{B + \sqrt{B^2 + 4A_{\mathrm{c}}P_{\mathrm{c}}m_{C_{\mathrm{T}}}D_{\mathrm{c}}}}, \quad a_{Ca^{2+}} = \frac{m_{\mathrm{Ca_T}}}{D_{\mathrm{c}} + P_{\mathrm{c}}a_{CO_3^{2-}}} \tag{10}$$

$B = A_{\mathrm{c}}D_{\mathrm{c}} + P_{\mathrm{c}}(m_{\mathrm{Ca_T}} - m_{C_{\mathrm{T}}})$; no Ca or no pairs gives $a_{CO_3^{2-}} = m_{C_{\mathrm{T}}}/A_{\mathrm{c}}$; $I \rightarrow 0$ gives ideal activities. c_{bulk} is the free Ca^{2+}. The local-ideal form $\Omega_{\mathrm{li}} = (c_{\mathrm{local}}/c_{\mathrm{bulk}})\Omega_{\mathrm{bulk}}$, $\Delta\mu_{\mathrm{li}} = \Delta\mu_{\mathrm{bulk}} - \Delta G_{\mathrm{seq}}/N_{\mathrm{A}}$, credits the binding free energy to the driving force; it is kept only as the named double-counting demonstration.

4 Classical nucleation (C3, R-slurry)

A cluster pays surface free energy before its volume gains $\Delta\mu$; the barrier is the top of that trade-off. On a substrate the cluster is a spherical cap with the same curvature.

$$\Delta G_{\mathrm{hom}}^* = \frac{16\pi\,\gamma_{\mathrm{nl}}^3 v_{\mathrm{m}}^2}{3\,\Delta\mu^2}, \quad r^* = \frac{2\gamma_{\mathrm{nl}}v_{\mathrm{m}}}{\Delta\mu}, \quad v_{\mathrm{m}} = \frac{V_{\mathrm{m}}}{N_{\mathrm{A}}} \quad [\mathrm{J};\ \mathrm{m}] \tag{11}$$

$$\Delta G_{\mathrm{het}}^* = f(\theta)\,\Delta G_{\mathrm{hom}}^*, \qquad f(\theta) = \tfrac{1}{4}(2+\cos\theta)(1-\cos\theta)^2 \tag{12}$$

$f(0) = 0$, $f(\pi/2) = \frac{1}{2}$, $f(\pi) = 1$; $f'(\theta) = \frac{3}{4}\sin^3\theta \ge 0$, so $0 \le \Delta G_{\mathrm{het}}^* \le \Delta G_{\mathrm{hom}}^*$ on $[0,\pi]$. $\Delta G_{\mathrm{hom}}^* \rightarrow \infty$ as $\Omega \rightarrow 1^+$, and $\Delta G_{\mathrm{hom}}^* \rightarrow 0$ as $\Omega \rightarrow \infty$.

5 Young, with one sign convention

θ is measured through the nucleus. Forming the contact removes substrate–liquid interface and creates substrate–nucleus interface over the same disc; $\cos\theta$ *is that energy gain per unit nucleus–liquid energy.*

$$\cos\theta = \frac{\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}}{\gamma_{\mathrm{nl}}}, \qquad \gamma_{\mathrm{sn}} = \gamma_{\mathrm{sn}}^{(0)} + \Delta\gamma_{\mathrm{w}} \quad [1;\ \mathrm{J\,m^{-2}}] \tag{13}$$

$\theta = 0$: the nucleus wets the template completely (no barrier); $\theta = \pi$: no wetting (homogeneous barrier). $\Delta\gamma_{\mathrm{w}} \equiv \gamma_{\mathrm{sn}} - \gamma_{\mathrm{sn}}^{(0)}$ is the free energy per unit contact area to displace the template-bound water layer; $\Delta\gamma_{\mathrm{w}} > 0$ **is a cost** and raises θ, f and $\Delta G_{\mathrm{het}}^*$ ($\partial\cos\theta/\partial\Delta\gamma_{\mathrm{w}} = -1/\gamma_{\mathrm{nl}} < 0$). If γ_{sn} is supplied as a full in-water interfacial free energy, $\Delta\gamma_{\mathrm{w}} \equiv 0$ and is not added again. $\gamma_{\mathrm{eff}} \equiv \gamma_{\mathrm{nl}} - \Delta\gamma_{\mathrm{w}}$ (deck form) is **not** used: Young and $\Delta G_{\mathrm{hom}}^*$ take the bare γ_{nl}. Reason: γ_{nl} multiplies the cap's free surface, which never touches the template, and $\Delta G_{\mathrm{het}}^* = f\,\Delta G_{\mathrm{hom}}^*$ follows only if the same γ_{nl} sits in Young and in $\Delta G_{\mathrm{hom}}^*$. With γ_{eff} in the denominator, $\partial\cos\theta/\partial\Delta\gamma_{\mathrm{w}} = (\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}^{(0)})/\gamma_{\mathrm{eff}}^2 > 0$ on the wetting side: a cost would lower the barrier.

6 Where the template acts (screen rule)

In the grand potential at the bulk chemical potentials every template–ion interaction sits in the interfacial energies, so the template acts only through θ; the driving force is the bulk one.

$$\Delta G_{\mathrm{het}}^* = f\big(\theta[\gamma_{\mathrm{sl}}^{(0)} + \Delta\gamma_{\mathrm{ads}},\, \gamma_{\mathrm{sn}}^{(0)},\, \Delta\gamma_{\mathrm{w}}]\big)\,\frac{16\pi\gamma_{\mathrm{nl}}^3 v_{\mathrm{m}}^2}{3\big(k_{\mathrm{B}}T\ln\Omega_{\mathrm{bulk}}\big)^2} \tag{14}$$

Rule: ΔG_{seq} enters once, through $\Delta\gamma_{\mathrm{ads}}$ in γ_{sl}; $\partial\Omega/\partial(\text{template inputs}) = 0$. If γ_{sl} is computed at the regime activities, add nothing. At fixed $\gamma_{\mathrm{sl}}^{(0)}, \gamma_{\mathrm{sn}}$, stronger binding *lowers* $\cos\theta$ and raises $\Delta G_{\mathrm{het}}^*$ (SF2): binding helps only if γ_{sn} falls

by more than γ_{sl}, i.e. if $\Gamma_{\mathrm{Ca,sn}} > \Gamma_{\mathrm{Ca,sl}}$. Lowering γ_{nl} lowers f only if $\gamma_{\mathrm{sl}} > \gamma_{\mathrm{sn}}$ (SF3). Proof: derivation, Section 7.

6a Accessibility of the barrier (C3)

Nuclei must form at the stated areal density within the time the template spends in the slurry and drying.

$$J = J_0 \mathrm{e}^{-\Delta G^*_{\mathrm{het}}/k_{\mathrm{B}}T}, \qquad N_{\mathrm{nuc}} = J t_{\mathrm{p}} \geq N_{\mathrm{req}} \\ \iff \quad \Delta G^*_{\mathrm{het}} \leq B_{\mathrm{acc}} = k_{\mathrm{B}} T \ln(J_0 t_{\mathrm{p}}/N_{\mathrm{req}}) \tag{15}$$

7 Conformality (C4, R-slurry)

A barrier needs a continuous film; whether the mineral spreads or beads is a wetting balance, not a barrier height.

$$S_{\mathrm{spread}} = \gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} - \gamma_{\mathrm{nl}} = \gamma_{\mathrm{nl}}(\cos\theta - 1) \quad [\mathrm{J\,m^{-2}}] \tag{16}$$

$S_{\mathrm{spread}} \geq 0$: no finite θ ($\theta \to 0$, $f \to 0$), layer-by-layer growth. $S_{\mathrm{spread}} < 0$: $\theta > 0$, island growth; a film forms only on coalescence, at cap height $h_{\mathrm{c}} = \tan(\theta/2)/(2\sqrt{N_{\mathrm{s}}})$, and closes at $h_{\mathrm{cl}} = \sqrt{2} h_{\mathrm{c}} \leq d_{\max}$ iff $N_{\mathrm{s}} \geq \tan^2(\theta/2)/(2d^2_{\max})$ [m; $\mathrm{m^{-2}}$]. A small-θ bound and $S_{\mathrm{spread}} \geq 0$ are one criterion at different strictness ($S_{\mathrm{spread}} = \gamma_{\mathrm{nl}}(\cos\theta - 1)$); coalescence is the independent one. A low f does not imply coverage: $f(\pi/2) = \frac{1}{2}$ is still island growth. h_{c} is geometric; the kinetic critical coalescence thickness of the charter is h_{coal} (not computed in P0).

8 Stability (C5): screens, necessary not sufficient

(a) R-slurry, aqueous Pourbaix, per non-O, non-H atom [$\mathrm{J\,mol^{-1}}$]:

$$\Psi_\phi = \tfrac{1}{n_{\mathrm{M}}}\big[G_\phi - n_{\mathrm{O}}\mu_{\mathrm{H_2O}} - RT\ln 10\,(2n_{\mathrm{O}} - n_{\mathrm{H}})\,\mathrm{pH} \\ - (2n_{\mathrm{O}} - n_{\mathrm{H}} + Q)FE_{\mathrm{SHE}}\big] \tag{17}$$

$\Delta G_{\mathrm{pbx}} = \Psi_\phi - \Psi_{\mathrm{hull}} \geq 0$; zero means stable at (pH, E_{SHE}).

(b) R-cell, Li open, E_{Li} vs $\mathrm{Li^+/Li}$ [$\mathrm{J\,mol^{-1}}$; V]:

$$\mu_{\mathrm{Li}} = \mu^\circ_{\mathrm{Li}} - FE_{\mathrm{Li}}, \qquad \Phi_\phi = G_\phi - n_{\mathrm{Li}}\mu_{\mathrm{Li}} \tag{18}$$

Window $[E_{\mathrm{red}}, E_{\mathrm{ox}}]$: the E_{Li} over which ϕ lies on the Li-open hull.

9 Insulating film (C4 limitation, R-cell)

The mineral is an electronic insulator, so conformality cannot be bought with thickness; the film must still pass the collector current.

$$R_A = \rho_{\mathrm{e}} d \leq R_{A,\max} \;\Rightarrow\; d \leq R_{A,\max}/\rho_{\mathrm{e}} \quad [\Omega\,\mathrm{m^2};\ \mathrm{m}] \tag{19}$$

$$j = j_0 \mathrm{e}^{-2\kappa d}, \quad \kappa = \frac{\sqrt{2 m_{\mathrm{e}} \phi_{\mathrm{B}}}}{\hbar} \;\Rightarrow\; d \leq \frac{\ln(j_0/j_{\mathrm{req}})}{2\kappa} \quad [\mathrm{m}] \tag{20}$$

Symbols

generated from `theory/symbols.yaml`; Reg.: S = R-slurry, C = R-cell; – = dimensionless

Symbol	Meaning	SI unit	Reg.
Exact SI constants (kept symbolic)			
k_B	Boltzmann constant	$J\,K^{-1}$	S,C
N_A	Avogadro constant	mol^{-1}	S,C
R	molar gas constant	$J\,K^{-1}\,mol^{-1}$	S,C
F	Faraday constant	$C\,mol^{-1}$	S,C
e	elementary charge	C	S,C
$\hbar$	reduced Planck constant	J s	C
m_e	electron mass	kg	C
State variables			
T	absolute temperature of the regime	K	S,C
pH	pH (molal scale)	–	S
E_{SHE}	collector potential vs SHE	V	S
E_{Li}	collector potential vs Li^+/Li	V	C
Adhesion in water (C1)			
γ_{ow}	oxide-water interfacial free energy	$J\,m^{-2}$	S
γ_{pw}	polymer-water interfacial free energy	$J\,m^{-2}$	S
γ_{op}	oxide-polymer interfacial free energy in contact	$J\,m^{-2}$	S
γ_o	oxide surface free energy against vacuum	$J\,m^{-2}$	S
γ_p	polymer surface free energy against vacuum	$J\,m^{-2}$	S
γ_w	water surface free energy against its vapour	$J\,m^{-2}$	S
γ_n	nucleus surface free energy against vacuum (dry)	$J\,m^{-2}$	S
$W_{adh,wet}$	work of adhesion in water (Dupre)	$J\,m^{-2}$	S
$W_{adh,dry}$	work of adhesion against vacuum (comparator)	$J\,m^{-2}$	S
ΔF_{PMF}	PMF free energy, separated minus bound	J	S
A_c	fragment contact area	m^2	S
Sequestration (C2)			
ΔG_{seq}	Ca^{2+} transfer free energy, bulk to binding layer	$J\,mol^{-1}$	S
ΔG_{site}	transfer free energy to the deprotonated site	$J\,mol^{-1}$	S
pK_a	pKa of the coordinating group	–	S
$\mathcal{E}$	Ca^{2+} enrichment factor (dilute limit)	–	S
c_{bulk}	bulk Ca^{2+} concentration	$mol\,m^{-3}$	S
c_{local}	mean Ca^{2+} concentration in the layer	$mol\,m^{-3}$	S
c_{max}	saturation concentration of the layer	$mol\,m^{-3}$	S
Γ_s	areal binding-site density	$mol\,m^{-2}$	S
δ_s	binding-layer thickness	m	S
V_{site}	volume per binding site	m^3	S
a_{site}	footprint area of one binding site	m^2	S
ρ_L	volume density of coordinating groups	$mol\,m^{-3}$	S
d_{layer}	template layer thickness	m	S
r_{Ca}	Ca^{2+} ionic radius	m	S
x	Langmuir binding variable	–	S
Θ	fraction of sites occupied by Ca^{2+}	–	S
Γ_{Ca}	Ca^{2+} surface excess (Gibbs)	$mol\,m^{-2}$	S
$\Gamma_{Ca,sn}$	Ca^{2+} excess at the template-nucleus contact	$mol\,m^{-2}$	S
Carbonate speciation (activities)			
m°	standard molality	$mol\,kg^{-1}$	S
ρ_w	density of water	$kg\,m^{-3}$	S
I	ionic strength (molal)	$mol\,kg^{-1}$	S
C_T	dissolved inorganic carbon	$mol\,m^{-3}$	S
Ca_T	total dissolved calcium	$mol\,m^{-3}$	S
pK_1	pK1 of carbonic acid	–	S
pK_2	pK2 of carbonic acid	–	S
$K_{CaCO_3^0}$	formation constant of $CaCO_3$(aq)	–	S
$K_{CaHCO_3^+}$	formation constant of CaHCO3+	–	S
${}^*K_{CaOH^+}$	hydrolysis constant of CaOH+	–	S
A_{DH}	Debye-Hueckel constant	–	S
b_D	Davies linear coefficient	–	S
I_{val}	upper validity bound of the activity model	$mol\,kg^{-1}$	S
$\gamma_{\pm1}$	activity coefficient, charge one	–	S
$\gamma_{\pm2}$	activity coefficient, charge two	–	S
m_{C_T}	dissolved inorganic carbon, molal	–	S
m_{Ca_T}	total calcium, molal	–	S
A_c	carbonate speciation factor	–	S
P_c	ion-pair factor	–	S
D_c	free-calcium factor	–	S
Supersaturation			
$a_{Ca^{2+}}$	bulk Ca^{2+} activity	–	S
$a_{CO_3^{2-}}$	bulk carbonate activity	–	S
K_{sp}	solubility product of the registered framing	–	S
a_w	water activity of the regime state	–	S
n	structural water per formula unit (hydrated framing)	–	S
Ω_{bulk}	bulk saturation ratio IAP/Ksp	–	S
Ω	saturation ratio where the nucleus forms	–	S
$\Delta\mu$	driving force per formula unit	J	S

Symbol	Meaning	SI unit	Reg.
Nucleation and Young (C3)			
γ_{nl}	nucleus-liquid interfacial free energy	$J\,m^{-2}$	S
γ_{sl}	template-liquid energy at the regime activities	$J\,m^{-2}$	S
$\gamma_{sl}^{(0)}$	template-liquid energy without Ca^{2+} binding	$J\,m^{-2}$	S
$\Delta\gamma_{ads}$	Ca^{2+} adsorption change of the template-liquid energy	$J\,m^{-2}$	S
γ_{sn}	template-nucleus energy in water (= ref + correction)	$J\,m^{-2}$	S
$\gamma_{sn}^{(0)}$	template-nucleus energy, reference without water displacement	$J\,m^{-2}$	S
$\Delta\gamma_w$	water-displacement correction (cost > 0)	$J\,m^{-2}$	S
γ_{eff}	deck effective nucleus-liquid energy (audit only)	$J\,m^{-2}$	S
θ	contact angle, measured through the nucleus	rad	S
$f(\theta)$	heterogeneous shape factor	–	S
V_m	molar volume of the registered framing	$m^3\,mol^{-1}$	S
r^*	critical radius	m	S
ΔG^*_{hom}	homogeneous nucleation barrier per nucleus	J	S
ΔG^*_{het}	heterogeneous barrier on the template	J	S
Accessibility of the barrier (C3)			
J_0	heterogeneous nucleation prefactor per area	$m^{-2}\,s^{-1}$	S
t_p	time available for nucleation	s	S
N_{req}	required areal density of nuclei	m^{-2}	S
A_{patch}	particle-collector contact patch	m^2	S
B_{acc}	largest accessible barrier	J	S
N_{nuc}	expected areal density of nuclei	m^{-2}	S
Conformality (C4)			
S_{spread}	spreading coefficient	$J\,m^{-2}$	S
N_s	island density at coalescence	m^{-2}	S
h_c	geometric coalescence thickness	m	S
h_{cl}	geometric closure thickness	m	S
h_{coal}	kinetic critical coalescence thickness (charter C4 (c))	m	S
Stability (C5)			
G_ϕ	molar Gibbs energy of species phi	$J\,mol^{-1}$	S
μ_{H_2O}	chemical potential of water	$J\,mol^{-1}$	S
n_O	number of O atoms per formula unit	–	S
n_H	number of H atoms per formula unit	–	S
n_M	number of non-O, non-H atoms	–	S
Q	species charge in units of e	–	S
Ψ_ϕ	Pourbaix potential per metal atom	$J\,mol^{-1}$	S
Ψ_{hull}	lower envelope of Psi (the hull)	$J\,mol^{-1}$	S
ΔG_{pbx}	energy above the aqueous Pourbaix hull	$J\,mol^{-1}$	S
n_{Li}	number of Li atoms per formula unit	–	C
μ°_{Li}	Li chemical potential in Li metal	$J\,mol^{-1}$	C
μ_{Li}	Li chemical potential at the electrode potential	$J\,mol^{-1}$	C
Φ_ϕ	Li-open grand potential	$J\,mol^{-1}$	C
ΔG_r	reaction free energy against Li metal	$J\,mol^{-1}$	C
Δn	Li released by the reaction	–	C
$\Delta\Phi$	grand-potential change of the reaction	$J\,mol^{-1}$	C
E^*	potential limit set by one reaction	V	C
E_{red}	lower limit of the stability window	V	C
E_{ox}	upper limit of the stability window	V	C
Insulating film (C4, R-cell)			
d	film thickness	m	C
ρ_e	film electronic resistivity	$\Omega\,m$	C
R_A	area-specific film resistance	$\Omega\,m^2$	C
$R_{A,max}$	admissible area-specific resistance (input)	$\Omega\,m^2$	C
ϕ_B	tunnelling barrier height	J	C
κ	tunnelling decay constant	m^{-1}	C
j_0	tunnelling prefactor current density	$A\,m^{-2}$	C
j_{req}	required current density (input)	$A\,m^{-2}$	C
d^{tun}_{max}	thickness bound, direct tunnelling	m	C
d_{max}	admissible film thickness	m	C
d^{ohm}_{max}	thickness bound, ohmic limit	m	C
Registration constants (theory/thresholds.md)			
p	credible level of the decision rule	–	S,C
p_{deg}	least mass shift that counts as degradation	–	S,C
m_W	C1 margin on the wet work of adhesion	$J\,m^{-2}$	S
m_{seq}	C2 margin on the Ca^{2+} transfer free energy	$J\,mol^{-1}$	S
m_{cos}	C3 margin on cos theta	–	S
m_S	C4 margin on the spreading coefficient	$J\,m^{-2}$	S
e_f	expected error of a computed formation energy	$J\,mol^{-1}$	S,C
n_{at}	atoms per formula unit	–	S,C
Δ_{ACC}	ACC offset above calcite	$J\,mol^{-1}$	S
τ_{hull}	C5 hull tolerance	$J\,mol^{-1}$	S
m_E	C5 margin on the window limits	V	C
E^{min}_{cell}	lower end of the R-cell potential range	V	C
E^{max}_{cell}	upper end of the R-cell potential range	V	C

Conventions. Angles in rad; free energies per formula unit (J) or per mole ($J\,mol^{-1}$) as stated; activities on the molal scale; potentials vs SHE in R-slurry and vs Li^+/Li in R-cell. Ω is the only supersaturation variable (the charter's S for C3 is Ω). The functions W4 implements, with exact limits, monotonicity and the sign-flip tests, are in `theory/contract.yaml`.